INTERNET OF THINGS

Question Bank

Unit-wise Topics | Easy, Medium & Hard Level Questions

Units I – V

UNIT I

Topic 1: Predecessors of IoT

EASY

1. What is the Internet of Things (IoT)?
2. Define Wireless Sensor Networks (WSN).
3. What is Machine-to-Machine (M2M) communication?
4. What is a Cyber Physical System (CPS)?
5. List the predecessors of IoT.

MEDIUM

1. Compare WSN and M2M communication.
2. Explain the role of Cyber Physical Systems in IoT.
3. How do WSNs enable IoT applications?
4. Explain the architecture of a Wireless Sensor Network.
5. Differentiate IoT and CPS.

HARD

1. Explain how WSN, M2M, and CPS evolved into IoT.
2. Design a smart agriculture system using WSN and CPS.
3. Analyze the limitations of M2M compared to IoT.
4. Explain how predecessors influenced modern IoT architecture.
5. Discuss the challenges faced before IoT emerged.

Topic 2: Emergence of IoT

EASY

1. Define IoT.
2. Explain the evolution of IoT.
3. What are IoT networking components?
4. What is an IP address in IoT?
5. What is IPv6?

MEDIUM

1. Explain enabling technologies of IoT.
2. Describe IoT networking architecture.
3. Explain addressing strategies in IoT.

4. Why is IPv6 preferred over IPv4 in IoT?
5. Explain the role of cloud computing in IoT.

HARD
1. Explain the complex interdependence of IoT technologies.
2. Compare centralized and distributed IoT networking.
3. Design an IoT architecture for a smart city.
4. Discuss the importance of addressing strategies.
5. Explain how IoT evolved from traditional Internet technologies.

UNIT II

Topic 1: IoT Sensing and Actuation

EASY

1. Define a sensor.
2. What is an actuator?
3. List different types of sensors.
4. Give examples of actuators.
5. What is sensing?

MEDIUM

1. Explain sensor characteristics.
2. Differentiate active and passive sensors.
3. Explain actuator characteristics.
4. Discuss different sensing types.
5. Explain sensor deviations.

HARD

1. Design a sensing system for smart irrigation.
2. Compare sensors and actuators.
3. Explain how sensors affect IoT accuracy.
4. Discuss challenges in selecting sensors.
5. Explain sensing considerations with examples.

Topic 2: IoT Processing Topologies

EASY

1. What is IoT processing?
2. Define data format.
3. What is edge processing?
4. What is cloud processing?
5. Why is processing required?

MEDIUM

1. Explain processing topologies.
2. Compare cloud and edge processing.
3. Explain processing offloading.

4. Describe IoT device selection considerations.
5. Explain data formatting techniques.

HARD
1. Design an IoT processing architecture.
2. Compare edge, fog, and cloud processing.
3. Discuss challenges in processing offloading.
4. Explain device selection for industrial IoT.
5. Evaluate processing strategies for smart healthcare.

UNIT III

Topic 1: IoT Connectivity Technologies

EASY

1. What is Zigbee?
2. Define LoRa.
3. What is Bluetooth?
4. What is Wi-Fi?
5. What is RFID?

MEDIUM

1. Compare Zigbee and Bluetooth.
2. Explain IEEE 802.15.4.
3. Differentiate NFC and RFID.
4. Explain LoRa architecture.
5. Explain Sigfox technology.

HARD

1. Compare LoRa, Sigfox, and NB-IoT.
2. Explain advantages of Thread protocol.
3. Design connectivity for a smart city.
4. Discuss industrial applications of WirelessHART.
5. Compare all IoT connectivity technologies.

Topic 2: IoT Communication Technologies

EASY

1. What is MQTT?
2. What is CoAP?
3. Define infrastructure protocol.
4. What is a discovery protocol?
5. What is device management?

MEDIUM

1. Explain MQTT architecture.
2. Compare MQTT and CoAP.
3. Explain semantic protocols.

4. Discuss data protocols.
5. Explain device management protocols.

HARD
1. Design communication architecture for healthcare IoT.
2. Explain protocol selection criteria.
3. Compare REST and MQTT.
4. Explain identification protocols with examples.
5. Discuss interoperability challenges.

UNIT IV

Topic 1: IoT Interoperability

EASY

1. Define interoperability.
2. What is an IoT standard?
3. Name IoT frameworks.
4. Why are standards important?
5. What is protocol interoperability?

MEDIUM

1. Explain interoperability challenges.
2. Compare IoT frameworks.
3. Explain IoT standards.
4. Discuss semantic interoperability.
5. Explain interoperability layers.

HARD

1. Design an interoperable IoT system.
2. Compare major IoT standards.
3. Discuss interoperability in healthcare.
4. Explain challenges of heterogeneous devices.
5. Evaluate interoperability solutions.

Topic 2: Fog Computing

EASY

1. Define Fog Computing.
2. What is edge computing?
3. List Fog Computing applications.
4. What is Fog architecture?
5. Why is Fog used?

MEDIUM

1. Explain Fog Computing architecture.
2. Compare Fog and Cloud.
3. Explain Fog in IoT.

4. Describe Fog applications.
5. Explain Fog benefits.

HARD
1. Design a Fog-based IoT system.
2. Compare Cloud, Fog, and Edge.
3. Explain latency reduction using Fog.
4. Discuss Fog security.
5. Explain real-time applications of Fog.

UNIT V

Topic 1: Paradigms, Challenges and Future

EASY

1. Define IoT paradigm.
2. List IoT challenges.
3. What is Industry 4.0?
4. What is Smart City?
5. What is Digital Twin?

MEDIUM

1. Explain IoT security challenges.
2. Explain IoT scalability.
3. Describe future IoT trends.
4. Explain emerging IoT pillars.
5. Discuss privacy issues.

HARD

1. Explain future IoT paradigms.
2. Design a secure IoT framework.
3. Discuss AI integration with IoT.
4. Explain blockchain in IoT.
5. Evaluate future challenges.

Topic 2: IoT Case Studies

Topic 2: Agricultural IoT

EASY

1. What is Vehicular IoT?
2. What is V2V communication?
3. What is V2I communication?
4. Give examples of Vehicular IoT.
5. Why is IoT used in transportation?

MEDIUM

1. Explain connected vehicles.
2. Describe intelligent transportation systems.

3. Explain smart traffic management.
4. Discuss vehicle tracking.
5. Explain accident detection systems.

HARD
1. Design a Vehicular IoT architecture.
2. Explain autonomous vehicle communication.
3. Discuss challenges in Vehicular IoT.
4. Compare V2V and V2I.
5. Explain future intelligent transportation systems.